

ThermoProt: Thermodynamically Motivated Flow Matching for Intrinsically Disordered Protein Ensembles

Anonymous Course Project Draft

Chemical Thermodynamics / Machine Learning for Molecular Ensembles

Abstract

Intrinsically disordered proteins (IDPs) require ensemble descriptions rather than single-structure prediction. We present ThermoProt, a research prototype that transports Gaussian, freely-jointed-chain, or Flory-inspired $C\alpha$ polymer priors toward data-derived IDP ensembles with optimal-transport conditional flow matching and an equivariant geometric network. Repository artifacts show promising local protocol results on IDRome20 and R_g -centered IDPForge S1/CALVADOS3 evaluations, while PED and NMR chemical-shift validation remain mixed. We therefore position ThermoProt as a thermodynamically motivated $C\alpha$ ensemble generator, not a calibrated equilibrium simulator or broad SOTA system.

1 Introduction

Classical protein modeling often asks for a single native structure, but many IDPs perform function through a distribution of rapidly interconverting conformations. For such systems, the object of interest is not a minimum-energy coordinate set but a probability measure over molecular microstates. This viewpoint is familiar in chemical thermodynamics: macroscopic observables arise from ensemble averages, and equilibrium distributions over chosen coarse variables can be idealized as Boltzmann weights after unresolved degrees of freedom have been integrated out.

ThermoProt approaches IDP ensemble generation from this perspective. Instead of starting from a black-box noise distribution alone, it can start from Gaussian, freely-jointed-chain (FJC), or Flory-inspired polymer priors. A neural velocity field then transports prior samples into data-like ensembles. The resulting system is not a molecular dynamics force field and does not currently provide a physical energy function. Its scientific value is narrower and more concrete: it offers a conceptual and computational bridge between polymer statistical thermodynamics and sequence-conditioned geometric deep learning, not a strict thermodynamic derivation.

Figure 1 summarizes the intended workflow. The paper asks how polymer statistical-thermodynamic priors can be introduced into IDP ensemble generation, and what the current local evidence can and cannot support.

We make three contributions. First, we formulate ThermoProt as a coarse-grained $C\alpha$ ensemble generator whose source distribution can encode simple polymer statistics. Second, we document a sequence-conditioned geometric flow model for transporting those priors toward data-derived IDP ensembles. Third, we audit the available local evidence and separate supported protocol-specific performance claims from limitations that require further validation.

2 Thermodynamic Framing

Let s denote a protein sequence, c optional experimental or environmental conditions, and $x \in \mathbb{R}^{N \times 3}$ a $C\alpha$ conformation. Because x contains only coarse $C\alpha$ coordinates, a thermody-

namic interpretation requires an effective potential of mean force (PMF) after marginalizing solvent, side-chain, and fast internal degrees of freedom. In that idealized setting, one would write

$$p^*(x | s, c) = Z(s, c)^{-1} \exp[-\beta W(x; s, c)], \quad (1)$$

where W is the coarse-grained PMF, $\beta = (k_B T)^{-1}$, and Z is the corresponding configurational partition function. ThermoProt does not learn W or Z explicitly. Instead, it learns a sampler whose output distribution is encouraged to match data-derived ensemble statistics.

For any structural or experimental observable $A(x)$, the thermodynamic quantity of interest is an ensemble average,

$$\langle A \rangle_{p^*} = \int A(x) p^*(x | s, c) dx. \quad (2)$$

This statement is useful even when the true PMF is unavailable: it clarifies that a generative model should be judged by distributions of observables rather than by a single representative conformation. It also clarifies the limitation of a purely coordinate-level loss. Matching sampled conformations can improve observable agreement, but it does not automatically calibrate relative free energies, barrier heights, or responses to changes in T , pH, salt, or solvent quality.

The most direct classical connection is polymer scaling. The radius of gyration,

$$R_g(x) = \left(\frac{1}{N} \sum_{i=1}^N \|r_i - r_{\text{cm}}\|^2 \right)^{1/2}, \quad (3)$$

is a standard global size observable. Flory-style scaling writes $R_g \sim a N^\nu$, with $\nu = 1/2$ for ideal random walks and $\nu \approx 0.588$ for good-solvent self-avoiding chains (Flory, 1953; Rubinstein and Colby, 2003). In ThermoProt this scaling motivates source priors and auxiliary observables, but it is not treated as an exact law for every sequence. For finite heteropolymers such as IDPs, ν is an effective exponent affected by charge patterning, hydrophobicity, solvent quality, and finite-size effects.

2.1 Polymer Priors as Inductive Biases

The source distribution is where the classical thermodynamic analogy enters most concretely. A Gaussian-chain prior can be written as a quadratic connectivity energy,

$$q_0(x | s) \propto \exp \left[-\frac{3}{2b^2} \sum_{i=1}^{N-1} \|r_{i+1} - r_i\|^2 \right], \quad (4)$$

where b is an effective segment length. An FJC prior instead treats segment lengths as approximately fixed while allowing bond directions to fluctuate. A Flory-inspired prior uses

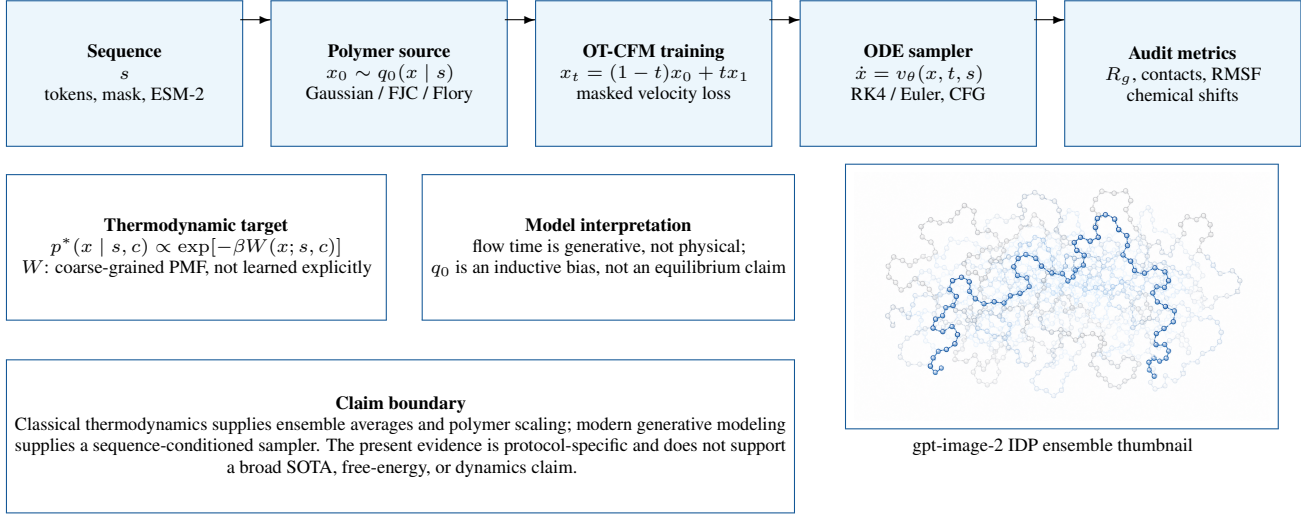


Figure 1. Publication schematic for ThermoProt. Deterministic L^AT_EX labels define the scientific workflow: sequence features condition a polymer source distribution, optimal-transport conditional flow matching learns a velocity field, ODE integration produces a $C\alpha$ ensemble, and downstream observables audit the result. The only generated raster component is the unlabeled ensemble thumbnail; it is illustrative and no quantitative value is read from it.

the qualitative balance between chain entropy and excluded-volume interactions,

$$\frac{F_{\text{Flory}}(R)}{k_B T} \approx \frac{3R^2}{2Nb^2} + \frac{vN^2}{R^3}, \quad (5)$$

to bias global compactness. These priors are useful because they encode zeroth-order polymer statistics before any sequence-conditioned neural transport is applied.

However, these priors should not be overinterpreted. They are not complete thermodynamic state equations for IDPs, and the Flory expression is a mean-field scaling argument rather than a sequence-specific PMF. In ThermoProt they serve as physically motivated source measures for a learned transport map. The scientific question is therefore whether the learned sampler improves ensemble observables while preserving sensible polymer-scale behavior, not whether it has derived the exact equilibrium distribution from first principles.

3 Related Work and Positioning

ThermoProt overlaps with recent ensemble generators but occupies a narrower point in the design space. STARLING uses a strong latent generative route and condition-aware IDR workflows (Novak et al., 2026). IDPFold and IDPForge pursue IDP-specific diffusion and all-atom-oriented generation (Zhu et al., 2025; Zhang et al., 2025). AlphaFlow applies flow matching to protein conformational landscapes (Jing et al., 2024). ThermoProt is closest to AlphaFlow in generative objective, but it emphasizes IDP polymer priors and coarse $C\alpha$ ensemble observables.

This positioning is deliberately modest. ThermoProt currently lacks explicit environmental conditioning, all-atom refinement, maximum-entropy reweighting, and a calibrated

free-energy surrogate. Repository readiness checks allow only dataset- and protocol-specific claims; they explicitly do not allow a broad SOTA claim.

4 Method

4.1 Conditional Flow Matching

ThermoProt trains a velocity model $v_\theta(x_t, t, s)$ using conditional flow matching (Lipman et al., 2023). Given a source sample $x_0 \sim q_0$ and a target conformation $x_1 \sim q_1$, a simple path is

$$x_t = (1 - t)x_0 + tx_1, \quad u_t = x_1 - x_0. \quad (6)$$

The model minimizes a squared velocity error, optionally with residue masks for variable-length proteins. Mini-batch optimal-transport coupling is used to pair source and target samples before interpolation, reducing arbitrary cross-pairing noise. The source q_0 may be a Gaussian chain or a polymer prior, giving the generative path a physically interpretable starting point. For a residue mask m_i , the implemented objective is a masked velocity loss,

$$\mathcal{L}_{\text{FM}} = \frac{\sum_i m_i \|v_\theta(x_t, t, s)_i - u_{t,i}\|_2^2}{3 \sum_i m_i + \epsilon}. \quad (7)$$

The interpolation time t is a generative training coordinate, not physical time. The path is not intended to represent molecular dynamics, a reversible thermodynamic transformation, or a minimum-free-energy pathway, and it does not by itself enforce detailed balance, excluded volume, or bond-length constraints.

At inference time, samples are generated by integrating

$$\frac{dx}{dt} = v_\theta(x, t, s) \quad (8)$$

with Euler, midpoint, Runge–Kutta, or adaptive ODE solvers. Classifier-free guidance is available through conditional/unconditional velocity interpolation, $v_{\text{cfg}} = (1 + w)v_\theta(x, t, s) - wv_\theta(x, t, \emptyset)$.

This objective has two practical advantages for a course-project system. It requires no explicit score estimate and no hand-designed reverse diffusion schedule, while still producing a continuous sampler that can be evaluated with different ODE solvers. The tradeoff is that CFM by itself optimizes transport toward the empirical target distribution used during training. It does not impose hard polymer constraints unless those constraints are built into the source distribution, architecture, loss, or post-processing checks. This is why the evaluation below treats bond geometry and observable closure as separate scientific requirements rather than assuming them from the training objective.

4.2 Geometric Architecture

The main network, GeoFormer, combines four ingredients:

- learned amino-acid and optional ESM-2 embeddings (Lin et al., 2023);
- Invariant Point Attention inspired by AlphaFold-style geometry processing (Jumper et al., 2021);
- E(n)-equivariant graph neural updates (Satorras et al., 2021);
- time-conditioned adaptive normalization and pair-distance features.

This design encodes a basic physical symmetry: translating or rotating an entire conformation should not change its molecular meaning. Dense pair features improve geometric reasoning but introduce an $O(N^2)$ cost, which is a central scaling limitation for long IDRs.

4.3 Ensemble Observables

The model is evaluated as a sampler, so observables are computed across a set of generated conformers $\{x^{(m)}\}_{m=1}^M$. The mean radius of gyration is

$$\overline{R_g} = \frac{1}{M} \sum_{m=1}^M R_g(x^{(m)}). \quad (9)$$

Contact maps are converted to probabilities using a distance cutoff d_c ,

$$P_{ij} = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{\|r_i^{(m)} - r_j^{(m)}\| < d_c\}, \quad (10)$$

and then compared with reference maps through Jaccard or divergence-style metrics. RMSF correlations summarize residue-wise positional variability after alignment or reference-frame normalization. Chemical-shift validation is more demanding because it requires approximate all-atom or local-geometry back-calculation from a C α ensemble; it is therefore treated as an external stress test rather than a direct training metric.

These observables deliberately cover different thermodynamic levels. R_g measures a one-dimensional global size distribution, contact maps probe nonlocal intrachain organization, RMSF probes residue-wise flexibility, and chemical shifts probe local conformational realism. Strong performance on one level does not imply success on the others. This hierarchy is central to the paper’s conservative claim language.

5 Experiments

All numbers in this paper are copied from repository-local JSON artifacts. They should be read as local protocol evidence, not as final benchmark claims. The stored target ensembles and observables combine simulated, benchmark, and baseline-derived artifacts depending on the dataset; each table therefore reports the protocol signal it supports rather than a universal ranking. These metrics primarily test ensemble geometry and selected observables. They do not by themselves validate an explicit thermodynamic potential, detailed balance, or calibrated environmental response.

5.1 Experimental Protocol

The canonical model for the main rerun tables is the epoch-75 vanilla v20 checkpoint with EMA disabled. IDRome and NMR use RK4-20 with classifier-free guidance 1.5; CALVA-DOS3 and PED use Euler-10 with guidance 1.0. IDRome matched reruns use STARLING DDIM-30 with the same conformer count as ThermoProt. PED historical rows mix unmatched baselines and are reported as a protocol audit, not a fair leaderboard. All tables should therefore be read together with dataset, checkpoint, sampler, guidance, conformer count, and homology/leakage status.

The local SOTA-readiness audit is used as a guardrail for claim language. It marks protocol-specific claims as allowed, but sets `broad_sota_allowed` to false because matched completion, timing, all-atom observable closure, IDPFold PED reruns, and v21 training evidence are incomplete. A separate condition-coverage audit also shows that the current benchmark manifests do not support environmental-response claims: IDRome is a simulation reference without experimental solution conditions, CALVADOS3 and IDPForge S1 do not report temperature, salt, pH, crowding, solvent-quality, or PTM values in the local manifest, and PED/BMRB contain only partial condition metadata. Consequently, the experiments test whether the sampler reproduces stored ensemble observables under fixed protocols; they do not test thermodynamic transfer across solution conditions.

5.2 IDRome 20 versus STARLING

Table 1 reports matched IDRome 20 comparisons against STARLING (Novak et al., 2026). ThermoProt improves R_g MAE, contact Jaccard, and cross-protein R_g correlation in these local matched runs. RMSF, contact JSD, and PCA-W2 are essentially tied, while STARLING is stronger on MI Spearman.

This result is promising but not sufficient for a broad SOTA claim. The IDRome 20 subset contains homology confounders in the local audit, and the original comparison used unequal conformer counts.

Table 1. IDRome 20 matched local comparison. ThermoProt uses epoch-75 vanilla, RK4-20, CFG 1.5; STARLING uses DDIM-30. Lower is better for R_g MAE, contact JSD, and PCA-W2; higher is better otherwise.

Metric	$n_{\text{conf}} = 50$		$n_{\text{conf}} = 100$	
	ThermoProt	STARLING	ThermoProt	STARLING
R_g MAE (Å)	1.436	1.719	1.445	1.860
RMSF corr.	0.915	0.916	0.918	0.924
Contact Jaccard	0.857	0.842	0.862	0.852
Contact JSD	0.055	0.055	0.053	0.053
PCA-W2	2337.0	2337.7	2336.9	2337.9
MI Spearman	0.003	0.060	-0.003	0.028
R_g Pearson r	0.965	0.948	0.968	0.938

Table 2. R_g benchmark summaries for fresh epoch-75 vanilla reruns. Error denotes mean corrected ΔR_g for IDPForge S1 and mean percent ΔR_g for CALVADOS3, so rows are not cross-dataset interchangeable.

Benchmark	n	n_{conf}	Mean R_g Err.	R_g MAE	$R_g r$
IDPForge S1	30	50	6.42%	3.01 Å	0.878
IDPForge S1	30	100	6.75%	3.13 Å	0.870
CALVADOS3	77	50	5.75%	1.98 Å	0.978
CALVADOS3	77	100	5.60%	1.93 Å	0.980

5.3 R_g -Centered Benchmarks

On the 30-protein IDPForge Table S1 benchmark (Zhang et al., 2025), fresh reruns with RK4-20 and guidance 1.5 give mean corrected R_g errors near 6–7%. On the 77-protein CALVADOS3 benchmark derived from the IDRome/CALVADOS route (IDRome Consortium, 2024), Euler-10 runs give mean R_g errors near 5.6%. These R_g -centered results validate a first-order global size observable; they do not prove the full PMF, local conformational statistics, or solvent response to be correct.

Checkpoint sweeps show tradeoffs rather than a single universally dominant model. Some EMA or later-epoch settings improve one scalar metric while degrading others. We therefore report checkpoint, sampler, guidance scale, and conformer count alongside benchmark results.

5.4 PED and NMR Mixed Results

The PED 80 evaluation is mixed. Historical ThermoProt variants improve R_g MAE relative to stored IDPFold and STARLING baselines under unmatched local protocols (Zhu et al., 2025), but IDPFold remains stronger on contact Jaccard and cross-protein R_g correlation. The v20 rerun improves contact Jaccard relative to older ThermoProt rows but has worse R_g MAE, indicating a compactness/shape tradeoff.

Chemical-shift validation is also mixed. With 50 conformers and SPARTA+-style back-calculation, nine proteins were evaluated. The all-set pooled χ^2/n is 5.06. Under the original leaked/clean split, seven clean proteins have pooled $\chi^2/n = 6.22$ and two leaked proteins have pooled $\chi^2/n = 2.74$. Under the more conservative manifest split, only one protein is clean (pooled $\chi^2/n = 4.34$) and eight are confounded ($\chi^2/n = 5.15$). Neither split supports a global NMR success claim.

Table 3. PED 80 protocol-separated local audit. Historical rows are not a matched leaderboard: IDPFold is $n=10$, STARLING is $n=50$, and the v20 row uses a fresh L256-bundle rerun with three truncated proteins.

Row	Protocol	R_g MAE	RMSF	Contact	$R_g r$
v13 FJC	historical	4.897	0.635	0.545	0.535
v11 Gaussian	historical	4.758	0.659	0.565	0.455
IDPFold	historical $n=10$	7.525	0.308	0.624	0.618
STARLING	baseline $n=50$	7.899	0.628	0.570	0.586
v20 rerun	fresh $n=50$	9.014	0.598	0.573	0.382

6 Discussion

6.1 What the Current Evidence Supports

The strongest current signal is that ThermoProt can generate plausible $C\alpha$ IDP ensembles whose global size statistics are competitive in several local protocols. The IDRome matched comparison is particularly useful because it controls conformer count against STARLING and reports several ensemble observables simultaneously. The CALVADOS3 and IDPForge S1 reruns show that the sampler can reproduce R_g -centered benchmarks at the few-percent level. Together, these results justify presenting ThermoProt as a thermodynamically motivated ensemble generator worth further study.

The evidence does not support a stronger interpretation. First, R_g is a single scalar and can be correct even when local chain organization is wrong. Second, contact and RMSF metrics are derived from coarse $C\alpha$ structures and do not fully test side-chain packing, hydrogen bonding, or solvent-mediated interactions. Third, the NMR chemical-shift audit remains difficult to interpret because clean/confounded splits change the apparent conclusion and because the back-calculation path is approximate. The right summary is therefore not “ThermoProt solves IDP thermodynamics”, but rather “ThermoProt supplies a coherent polymer-prior flow-matching baseline whose successes and failures can be audited with thermodynamic observables.”

6.2 SOTA Gap Analysis

Table 4 summarizes what would be required before the work could be presented as a broad SOTA system. The table is intentionally framed as a research agenda rather than as a claim of completion.

This gap analysis also explains why the chemical-thermodynamics framing is valuable. It gives a principled way to decide which future experiments matter: not only lower coordinate error, but also correct ensemble averages, response functions, uncertainty, and observable closure under controlled conditions. Modern generative modeling provides the computational route, while classical thermodynamics defines many of the questions that the route must answer.

7 Thermodynamic Extension Plan

For a chemical-thermodynamics course project, the most natural extension is not to claim that the neural sampler is a force field, but to use it as a platform for testing thermodynamic ideas on IDP ensembles. Four concrete modules follow from the current repository state.

Table 4. SOTA-readiness gaps implied by local audits and reviewer feedback.

Requirement	Current state	Needed evidence
Matched baselines	IDRome matched; PED historical rows mixed	Equal conformer counts, shared splits, rerun baselines
Experimental closure	NMR mixed; no SAXS, FRET, PRE, or RDC closure	Back-calculated observables with clean condition metadata
Thermodynamic calibration	Polymer priors but no learned PMF	Reweighting or free-energy surrogate validated by perturbations
Environmental response	Conditions largely absent or partial	Temperature, pH, salt, crowding, solvent-quality benchmarks
All-atom realism	$C\alpha$ generator only	Side-chain and all-atom refinement with steric validation
Scaling and speed	$O(N^2)$ pair features; timing not claim-ready	Long-IDR profiling and sparse or low-rank architectures

7.1 Polymer Scaling and Solvent-Quality Diagnostics

The first module is a systematic scaling analysis. For generated ensembles, one can fit

$$\log \overline{R_g} = \log a + \nu \log N \quad (11)$$

across proteins or sequence windows and compare the fitted ν with ideal-chain, good-solvent, and poor-solvent expectations. This should be done with bootstrap confidence intervals and with stratification by charge fraction, net charge per residue, hydropathy, and proline/glycine content. If the model uses a Flory-inspired prior, the fitted exponent provides a direct sanity check: the prior should influence the source distribution without forcing every sequence into the same universal compactness regime.

Beyond R_g , the same module can audit end-to-end distance, asphericity, bond lengths, and contact probability as a function of sequence separation $P_c(|i - j|)$. These are simple observables, but they are thermodynamically meaningful because they distinguish global coil size from local chain packing. A model that matches R_g while producing unphysical bond lengths or contact decay would be exposed immediately.

7.2 Maximum-Entropy Reweighting

A second module is to add a lightweight thermodynamic correction layer after generation. Given samples from $q_\theta(x | s)$ and experimental observables a_k^{exp} , one can define a maximum-entropy reweighted ensemble,

$$q_\lambda(x | s) = \frac{q_\theta(x | s) \exp[-\sum_k \lambda_k A_k(x)]}{Z_\lambda(s)}, \quad (12)$$

where the multipliers λ_k are chosen so that $\langle A_k \rangle_{q_\lambda} \approx a_k^{\text{exp}}$. This is a standard statistical-thermodynamic idea: adjust the least biased distribution consistent with measured observables. In the ThermoProt setting, the base sampler supplies candidate

conformations and the reweighting layer supplies a transparent way to incorporate SAXS, FRET, PRE, or chemical-shift constraints without retraining the neural network.

This extension would also make the limitations measurable. If only extreme weights can match an experiment, then the generated ensemble lacks support in the relevant conformational region. If modest weights close the observable gap, then the sampler may already contain plausible conformations but needs experimental calibration. Effective sample size, KL divergence from q_θ , and observable residuals would become interpretable diagnostics.

7.3 Response Functions and Condition Dependence

The current repository does not yet support condition-dependent thermodynamic claims, but it can be extended in that direction. In equilibrium statistical mechanics, response functions connect observable changes to fluctuations. If a condition variable η couples to a coarse observable $B(x)$, then an idealized response relation has the covariance form

$$\frac{\partial \langle A \rangle}{\partial \eta} \propto -\beta \text{Cov}(A, B). \quad (13)$$

For IDPs, η might represent salt concentration, pH, solvent quality, or crowding in a learned conditional model. A future ThermoProt benchmark could therefore evaluate not only whether $\overline{R_g}$ is correct at one condition, but whether the direction and magnitude of compaction or expansion under a perturbation are physically plausible.

This is where modern thermodynamics becomes relevant. The flow model is a non-equilibrium transport map in generative time, not a physical process, but its learned distribution can still be probed with equilibrium-style observables and response tests. A successful system would connect the two viewpoints: a fast neural sampler that respects symmetries and polymer statistics, plus thermodynamic diagnostics that prevent unsupported extrapolation.

8 Limitations

ThermoProt should not be interpreted as an equilibrium simulator. It does not produce a molecular dynamics trajectory, enforce microscopic reversibility, or provide an absolute free energy. The ODE integration variable is a model time, not a physical time. The current $C\alpha$ representation omits side-chain, solvent, ion, and all-atom steric details; chemical shifts and other local observables therefore require approximate back-calculation and should be treated as validation probes rather than training guarantees.

The empirical evidence is also limited. IDRome comparisons are local and contain homology caveats. PED results are not uniformly favorable. NMR chemical-shift validation is mixed, with clean-set aggregate error still high. Finally, the current paper uses a local AAAI-style fallback template; official AAAI formatting files should replace it before any real submission.

9 Future Work

The next scientific step is to move toward thermodynamic calibration. This requires conditioning on temperature, pH,

salt, solvent quality, crowding, or post-translational modifications, and validating how those conditions change ensemble observables. A direct course-project extension is to fit $R_g \sim N^\nu$ and contact-order scaling across generated ensembles, then compare the inferred exponents with ideal-chain, self-avoiding-chain, and poor-solvent expectations. Additional sanity checks should include $C\alpha$ bond lengths, end-to-end distance, asphericity, excluded-volume violations, and contact probability as a function of sequence separation.

A second direction is to learn an implicit free-energy or reweighting model so that generated ensembles can be corrected against SAXS, FRET, PRE, or chemical-shift data with minimal distributional perturbation. A third direction is computational: dense pair features and triangle-style updates should be replaced or complemented by sparse or low-rank routes for long IDRs.

10 Conclusion

ThermoProt shows how ensemble thermodynamics can organize a modern neural generator for IDP conformations: polymer priors define a physically meaningful source, flow matching learns a distributional transport, and geometric networks respect rigid-motion symmetries. The current evidence is useful for a course project and for protocol-specific research discussion, but not for broad SOTA or strict thermodynamic-simulation claims.

References

- Paul J. Flory. *Principles of Polymer Chemistry*. Cornell University Press, 1953.
- IDRome Consortium. A proteome-scale atlas of intrinsically disordered protein conformations. *Nature*, 2024. URL <https://pubmed.ncbi.nlm.nih.gov/38297118/>.
- Bowen Jing, Bonnie Berger, and Tommi Jaakkola. Alphafold meets flow matching for generating protein ensembles. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. URL <https://proceedings.mlr.press/v235/jing24a.html>.
- John Jumper, Richard Evans, Alexander Pritzel, et al. Highly accurate protein structure prediction with alphafold. *Nature*, 596:583–589, 2021.
- Zeming Lin, Halil Akin, Roshan Rao, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379:1123–1130, 2023.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- Borna Novak, Jeffrey M. Lotthammer, Ryan J. Emenecker, and Alex S. Holehouse. Accurate predictions of disordered protein ensembles with starling. *Nature*, 2026. doi: 10.1038/s41586-026-10141-2. URL <https://www.nature.com/articles/s41586-026-10141-2>.
- Michael Rubinstein and Ralph H. Colby. *Polymer Physics*. Oxford University Press, 2003.
- Victor Garcia Satorras, Emiel Hoogetboom, and Max Welling. E(n) equivariant graph neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, 2021.
- Oufan Zhang, Zi Hao Liu, Julie D. Forman-Kay, and Teresa Head-Gordon. Deep learning of proteins with local and global regions of disorder, 2025. URL <https://arxiv.org/abs/2502.11326>.
- J. Zhu, Z. Li, Z. Zheng, et al. Accurate generation of conformational ensembles for intrinsically disordered proteins with idpfold. *Advanced Science*, 12(48):e11636, 2025. doi: 10.1002/advs.202511636. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC12752595/>.